

University of Petroleum and Energy Studies

GHOSTEX

Smart Financial Tracker & Analytics Engine

Project Report

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1. Executive Summary

GHOSTEX is a robust, desktop-based financial application engineered to help users track their daily expenses, manage recurring subscriptions, and gain intelligent insights into their spending habits. Built entirely in Python, the system seamlessly integrates a modern graphical user interface (GUI) with a backend data management system and a mathematical analytics engine.

Project Objective: To build a completely reactive, state-managed desktop application that moves beyond basic data entry to provide proactive financial intelligence, including automated billing rollovers and mathematical spending anomaly detection.

2. System Architecture

The application is structured using professional software engineering principles, separating the interface, data management, data models, and analytical logic into distinct, modular components.

2.1 User Interface Layer (`main.py`)

Built using **CustomTkinter**, the UI provides a dark-mode, modern aesthetic. It utilizes realtime state management to ensure that whenever the underlying database changes (e.g., saving an expense or deleting a subscription), the UI instantly reflects these updates. Key UI components include:

- **Dynamic Ledgers:** Scrollable history frames with built-in, secure deletion functionality.
- **Real-Time Dashboard:** A reactive overview showing total spend, annual subscription costs, and an upcoming bills alert system.

- **Validation Bouncers:** Pre-save logic checks that prevent duplicate IDs or malformed data from corrupting the database.

2.2 Data Management Layer (`data_manager.py`)

A robust CSV-based database handler that manages data persistence. It utilizes standard Python library tools like `csv.DictReader` and `csv.DictWriter` to safely append, load, and rewrite data without corruption. It also houses the "Time-Travel Rollover" engine, which automatically recalculates past-due subscription dates upon application boot.

2.3 Analytical Engine (`analytics.py`)

The "Brain" of GHOSTEX. This module leverages high-performance data science libraries to process raw dictionaries from the DataManager. It calculates total expenditures, groups spending by category, and implements advanced statistical checks.

2.4 Object Models (`models.py`)

Provides Object-Oriented blueprints for `Expense` and `Subscription` records. It strictly enforces data validation upon instantiation (e.g., preventing negative amounts, ensuring correct date formatting) before handing the verified data back to the DataManager.

3. Key Features & Smart Capabilities

Feature	Description
Anomaly Detection	Utilizes NumPy to calculate the Mean and Standard Deviation of historical spending. Automatically flags any transaction that exceeds 1.5 standard deviations above the average as an "Unusual Spend".
Automated Rollover	On application boot, the system scans for past-due subscriptions and automatically pushes their next billing dates forward (Monthly, Weekly, or Yearly) based on their cycle.

State Synchronization	Every addition or deletion triggers a backend data reload and a UI refresh, ensuring the Dashboard and Ledgers are never displaying stale data.
Proactive Alerts	Filters subscriptions to calculate the difference between the current date and upcoming due dates, alerting the user to bills due within the next 3 days on the main dashboard.

4. Technology Stack

Python 3.x

CustomTkinter

Pandas

NumPy

CSV Module

Datetime

5. Conclusion

GHOSTEX successfully demonstrates the integration of a responsive desktop GUI with data processing and statistical analysis. By adhering to clean code architecture and eliminating derived data storage, the application remains lightweight, fast, and highly accurate. It serves as a fully functional proof-of-concept for a personal financial advisor application.